

Aryl hydrocarbon receptor literature

Bibliometric Analysis Summary Report

Example configuration for a conservative OpenAlex bibliometric pipeline focused on the aryl hydrocarbon receptor literature.

Query definition

OpenAlex exact-match title/abstract retrieval for "aryl hydrocarbon receptor", "ah receptor", and "dioxin receptor", plus title-level "AHR" hits; local regex validation retained records with explicit AhR naming in the title or abstract-phrase hits supported by biologically relevant title language.

Key corpus statistics

Validated papers	8,291
Time window	1980 to 2026
Abstract coverage	5,290 (63.8%)
Country metadata	7,380 (89.0%)
Disease/application tagged	7,231 (87.2%)
Focus-tagged papers	6,766 (81.6%)
Journals represented	1,609
Countries represented	107

Report scope

The following pages reproduce the generated thesis figures in order. This report is assembled from the current figure files without altering their styling or content.

Corpus Filtering and Retention Flow

Technical overview of retrieval, deduplication, validation, and metadata availability.

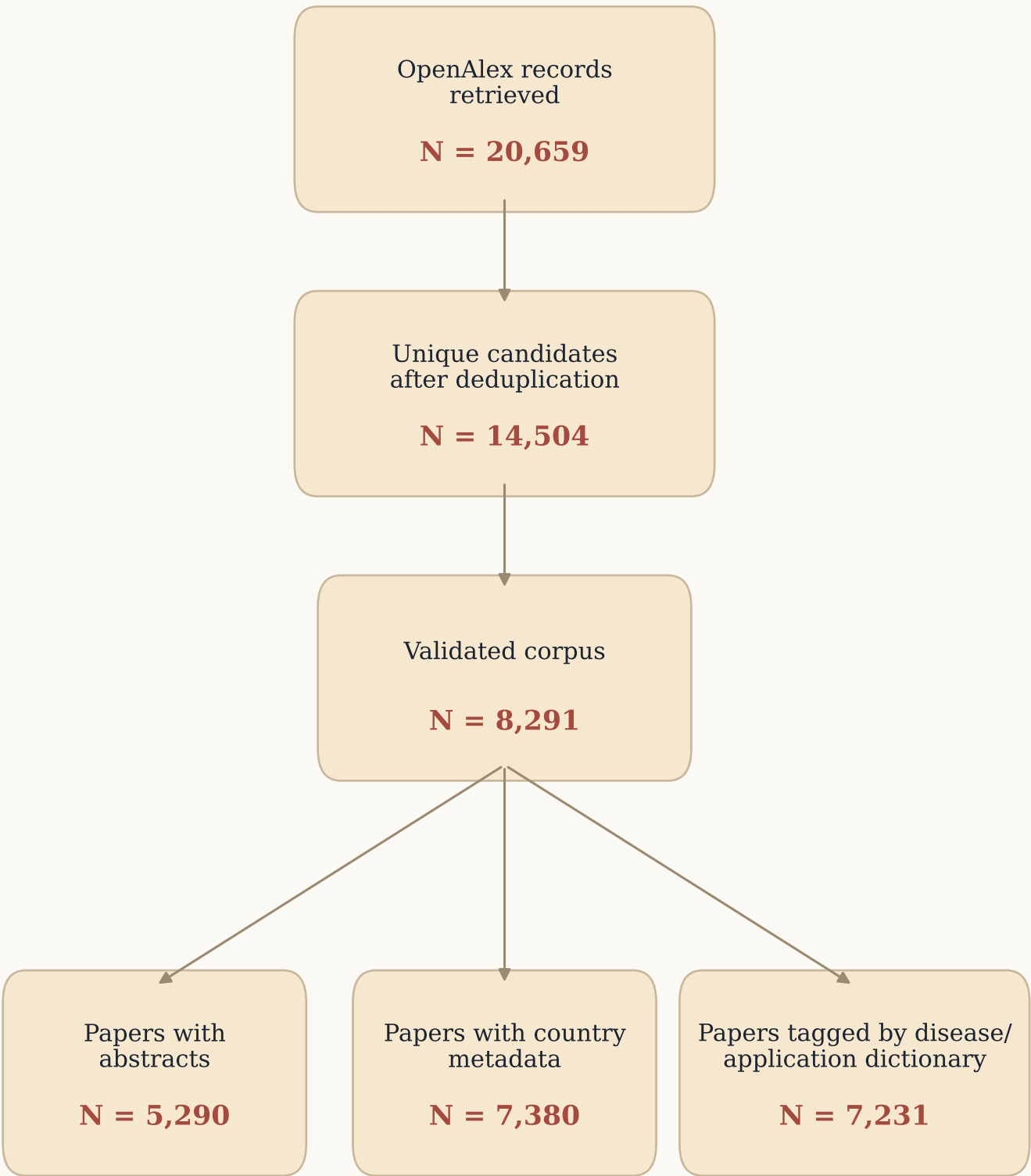
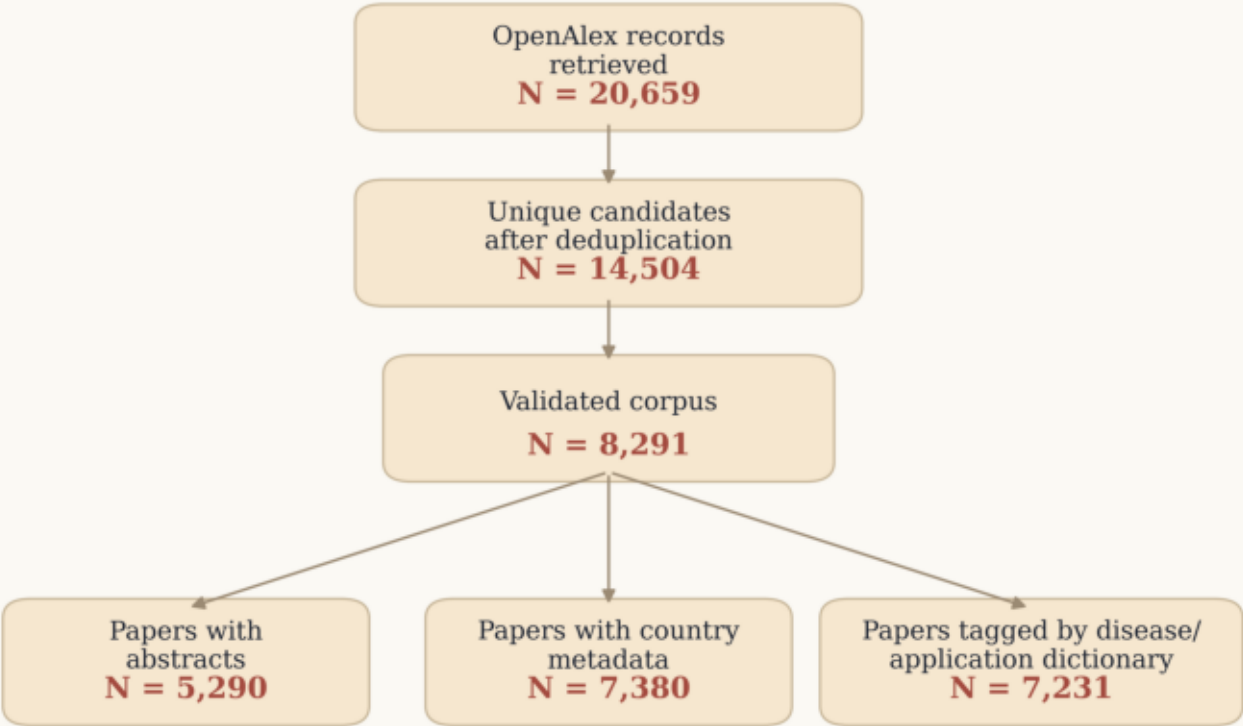


Figure 00. Corpus filtering and metadata retention flow

Corpus Filtering and Retention Flow

Technical overview of retrieval, deduplication, validation, and metadata availability.



Counts are drawn from the current cached retrieval, validated corpus table, and downstream metadata coverage summaries.

Figure 01. AhR literature growth over time

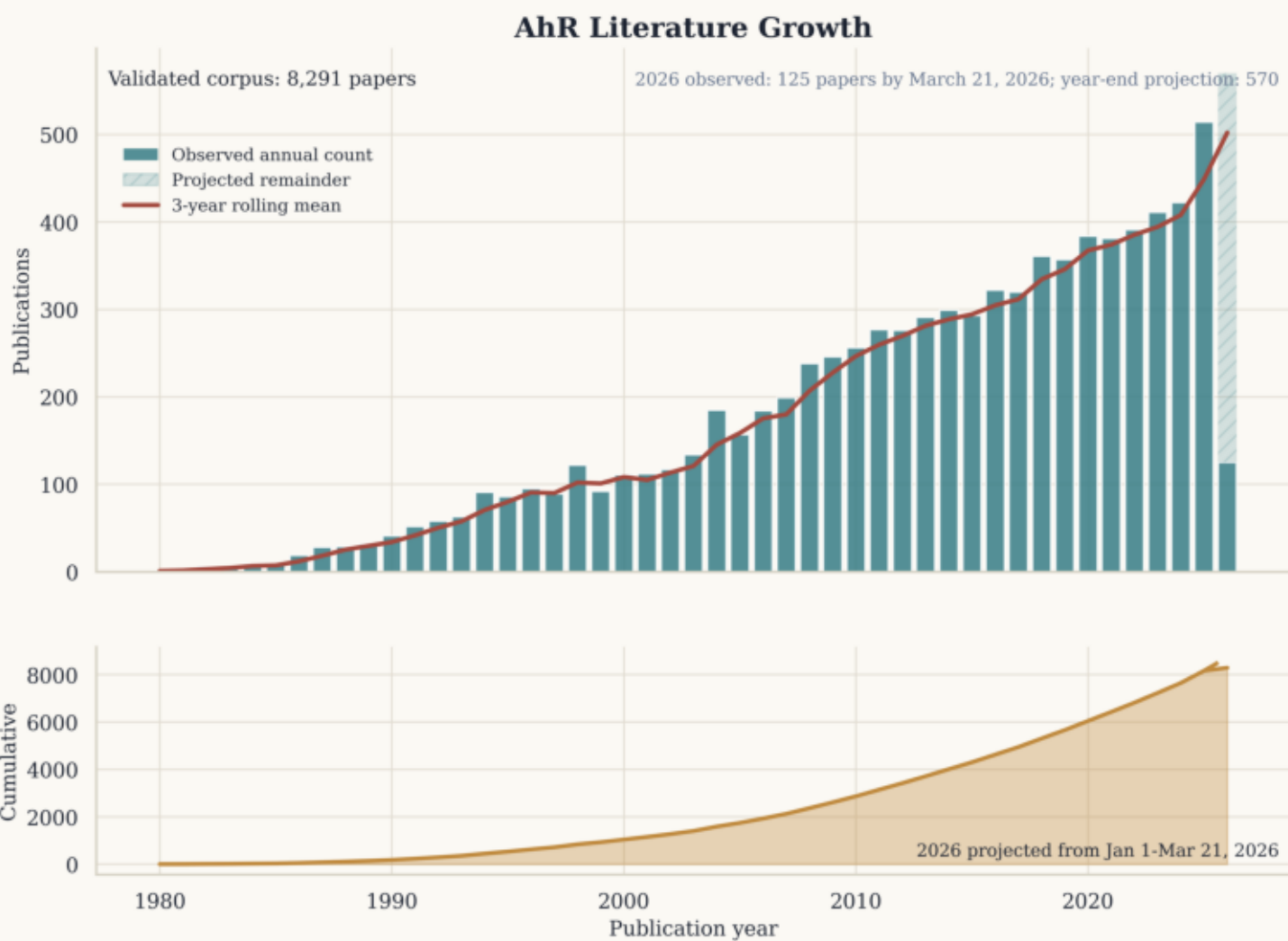


Figure 02. Disease and application distribution across the AhR corpus

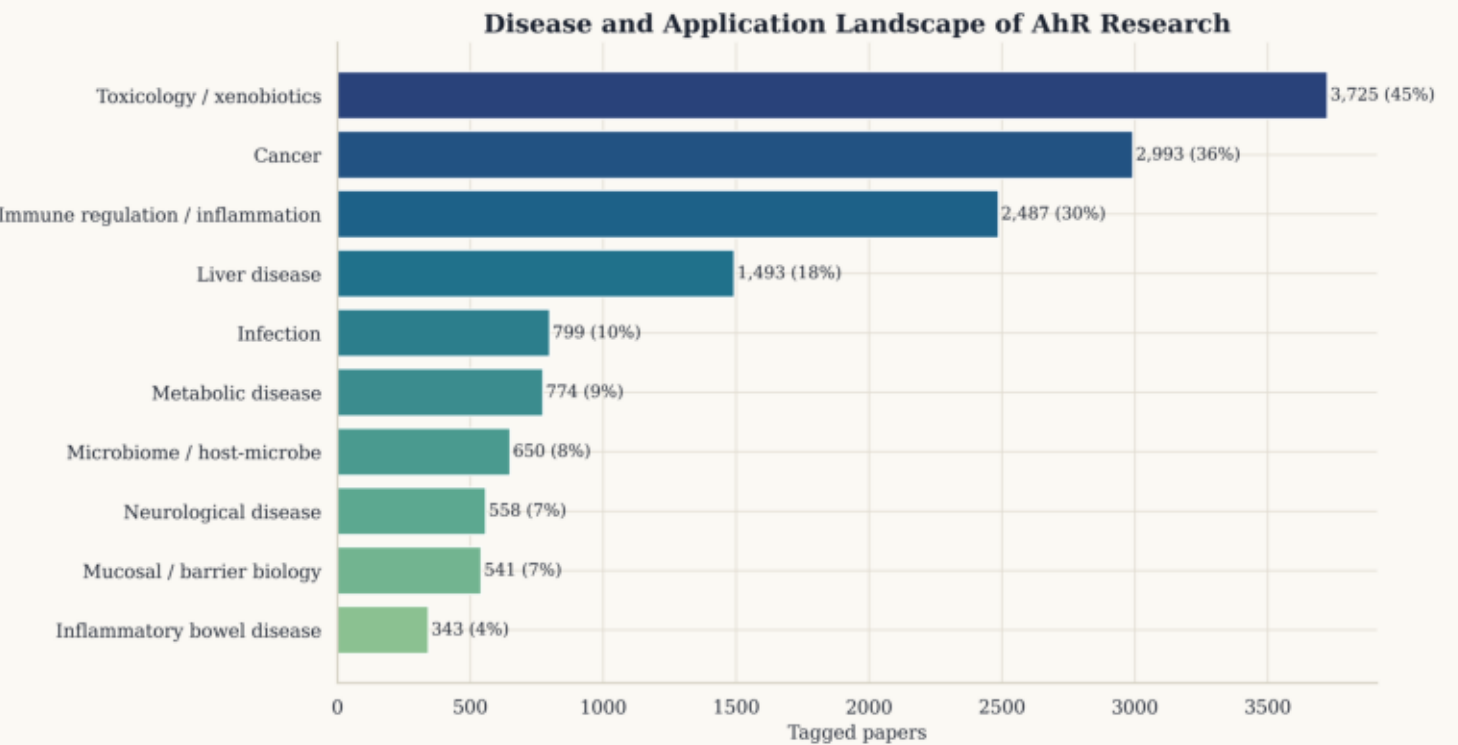
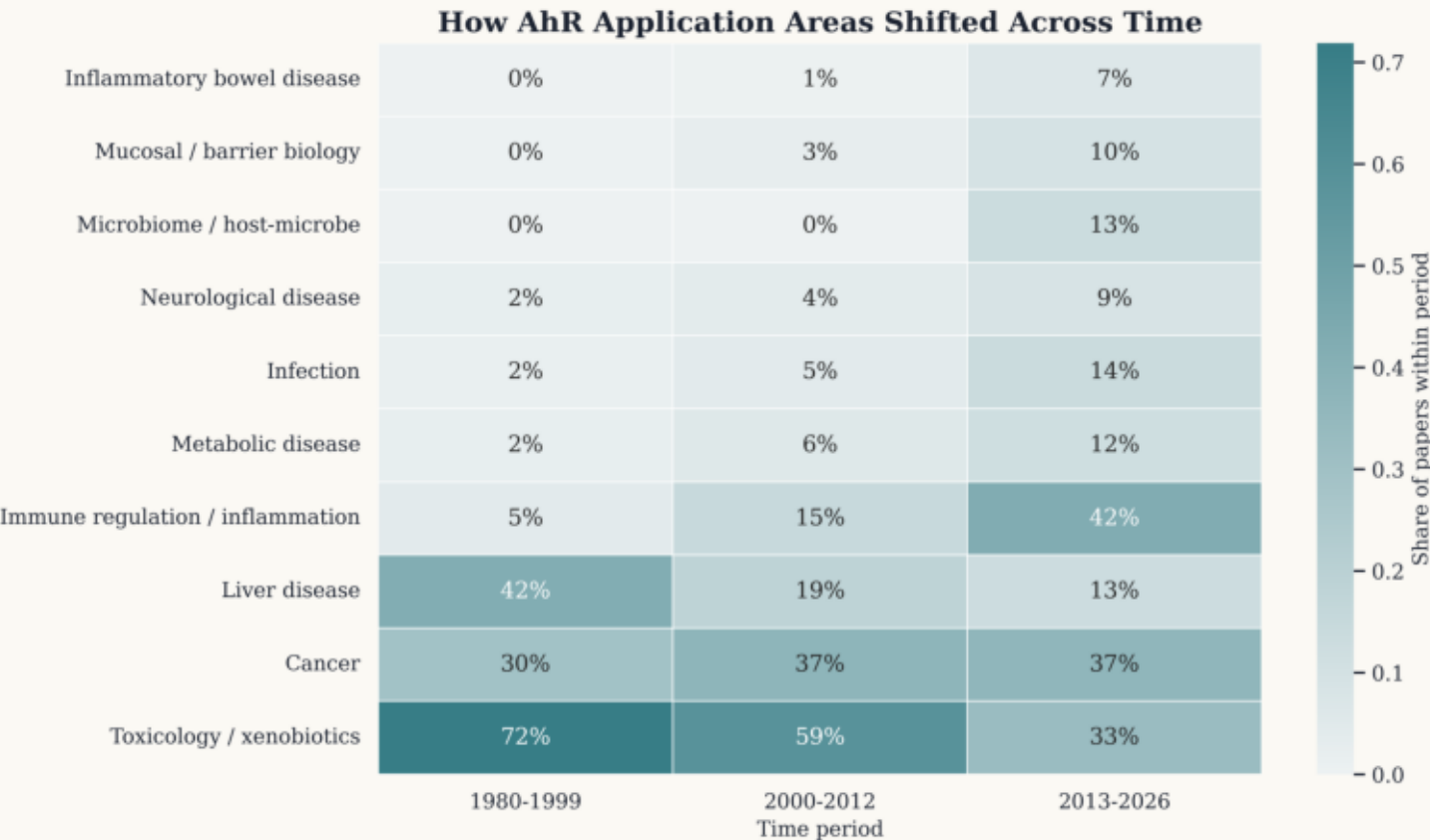


Figure 03. Disease and application trends across AhR field eras



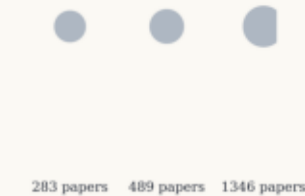
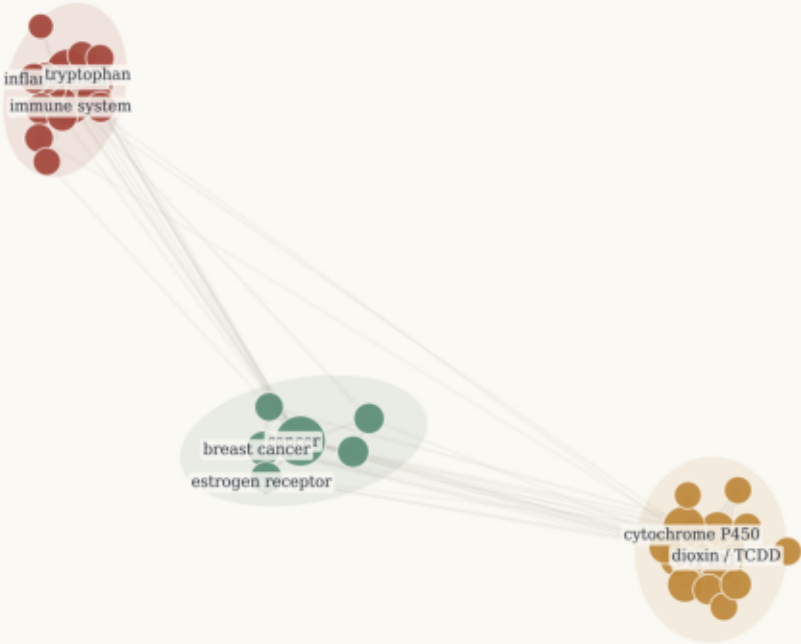
Conceptual Landscape of AhR Research

How to read this map

- Node size: number of papers carrying the concept
- Node color: thematic cluster from Louvain community detection
- Edge width: co-occurrence strength between concept labels

Conceptual Landscape of AhR Research

Curated concept co-occurrence map across the full validated AhR corpus



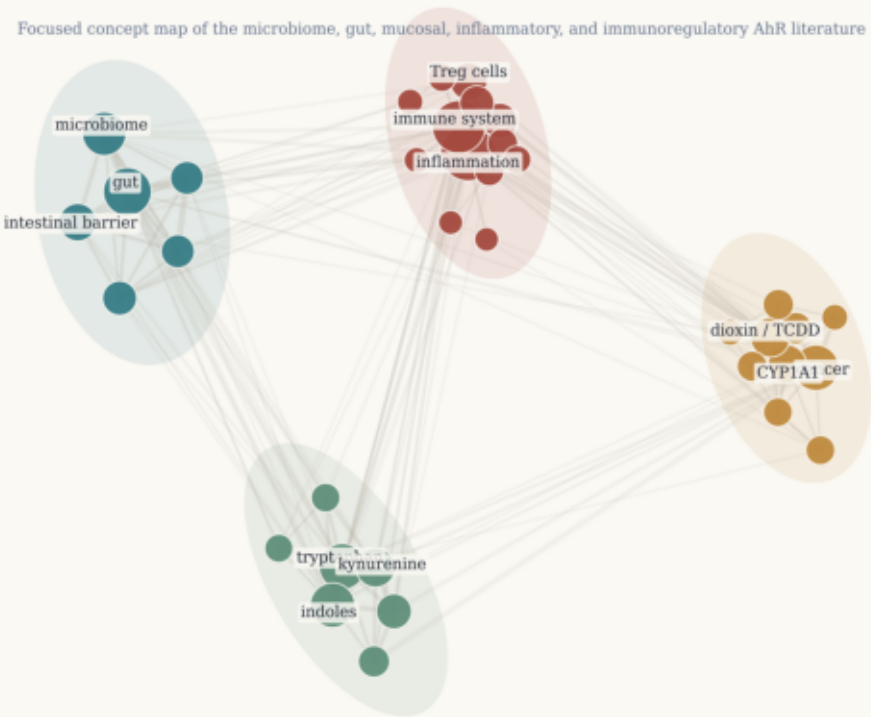
Thematic clusters

- Cluster 2: Dioxin and xenobiotic toxicology
CYP1A1, dioxin / TCDD, cytochrome P450, CYP1B1, carcinogen
- Cluster 1: Immune-microbiome signaling
immune system, inflammation, tryptophan, gut, indoles
- Cluster 3: Cancer and hormone signaling
cancer, breast cancer, estrogen receptor, carcinogenesis, oxidative stress

Immune-Microbiome-Barrier AhR Sublandscape

Immune-Microbiome-Barrier AhR Sublandscape

Focused concept map of the microbiome, gut, mucosal, inflammatory, and immunoregulatory AhR literature



How to read this map

Node size: number of papers carrying the concept

Node color: thematic cluster from Louvain community detection

Edge width: co-occurrence strength between concept labels



169 papers 249 papers 658 papers

Thematic clusters

- Cluster 1: T cell and cytokine regulation**
inflammation, immune system, Treg cells, TNF-alpha, Th17 cells
- Cluster 2: CYP1 toxicology and carcinogenesis**
cancer, CYP1A1, dioxin / TCDD, cytochrome P450, CYP1B1
- Cluster 4: Microbiome and gut barrier biology**
gut, microbiome, intestinal barrier, IL-22, inflammatory bowel disease
- Cluster 3: Tryptophan-kynurenine immunometabolism**
tryptophan, indoles, kynurenine, IDO1, tumor immunity

Thematic Evolution of AhR Research

Thematic Evolution of AhR Research

Ribbon width shows the share of papers assigned to each thematic cluster within each era.

1980-1999 2000-2012 2013-2026

928 papers

2,492 papers

4,871 papers

How to read this map

Ribbon color: thematic cluster
Ribbon width: within-period cluster share
Columns: 1980-1999, 2000-2012, 2013-2026

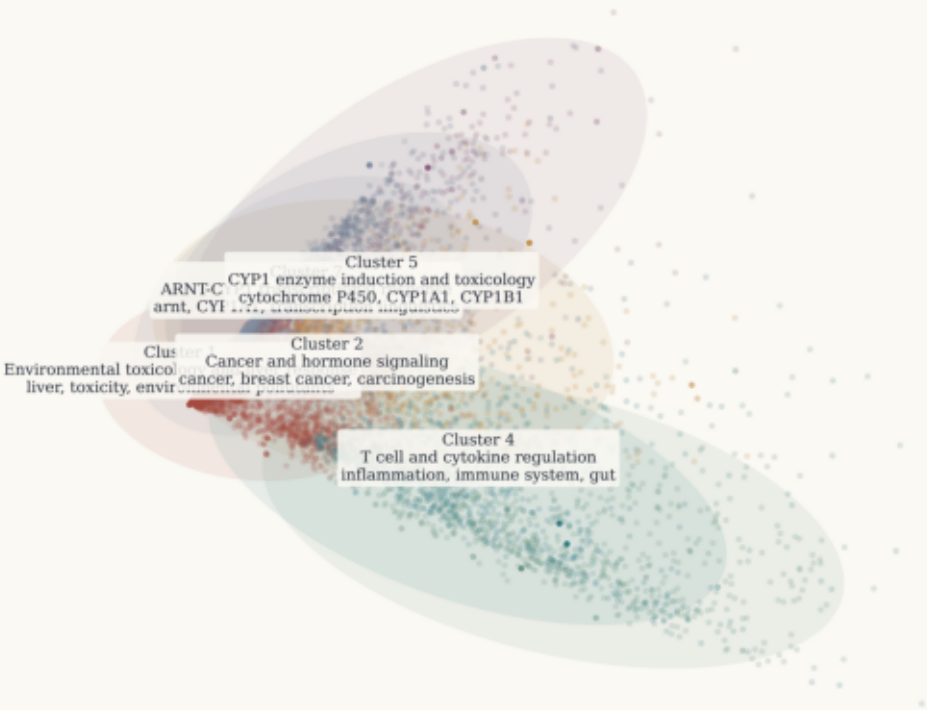
Clusters tracked across time

- Cluster 1: Environmental toxicology and liver response
Recent-era share: 38%
- Cluster 2: Cancer and hormone signaling
Recent-era share: 15%
- Cluster 7: ARNT-CYP1 transcriptional response
Recent-era share: 7%
- Cluster 5: CYP1 enzyme induction and toxicology
Recent-era share: 6%
- Cluster 4: T cell and cytokine regulation
Recent-era share: 15%
- Cluster 3: Immune-microbiome signaling
Recent-era share: 16%
- Cluster 6: CYP1 toxicology and carcinogenesis
Recent-era share: 2%

Document Landscape of AhR Research Themes

Document Landscape of AhR Research Themes

Each point is one paper positioned in 2D concept space; colored islands summarize the major thematic clusters.



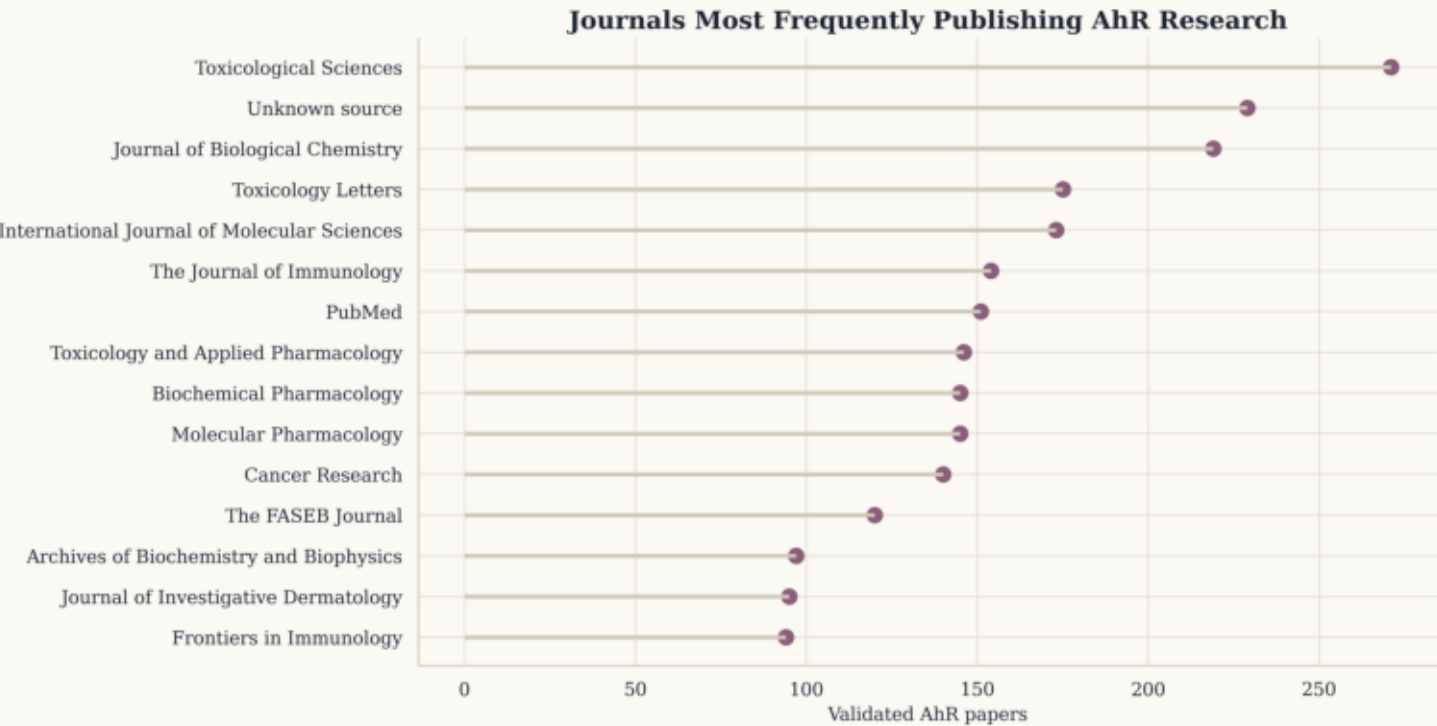
How to read this map

- Each point: one paper
- Color and island envelope: thematic cluster
- Distances: qualitative similarity in concept-profile space

Cluster summaries

- Cluster 1: Environmental toxicology and liver response
3,464 papers | median year 2013
- Cluster 2: Cancer and hormone signaling
1,039 papers | median year 2018
- Cluster 7: ARNT-CYP1 transcriptional response
976 papers | median year 2008
- Cluster 5: CYP1 enzyme induction and toxicology
884 papers | median year 2008
- Cluster 4: T cell and cytokine regulation
874 papers | median year 2020
- Cluster 3: Immune-microbiome signaling
803 papers | median year 2022
- Cluster 6: CYP1 toxicology and carcinogenesis
251 papers | median year 2011

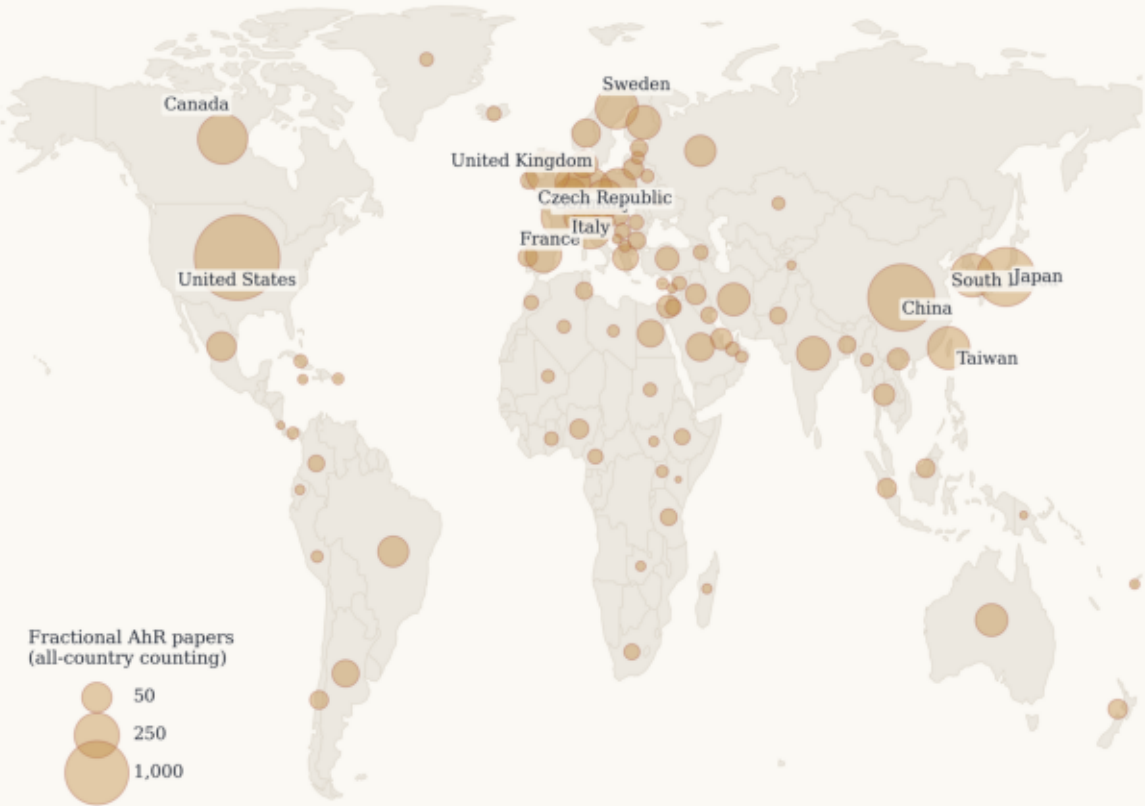
Figure 08. Journals most frequently publishing AhR papers



Global AhR Research Output

Global AhR Research Output

Bubble area shows fractional AhR paper count by country.



Global AhR Research Output Per Capita

Global AhR Research Output Per Capita

Bubble area shows fractional AhR papers per million inhabitants.

